

Crop Yield Prediction Using Machine Learning

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**DEPARTMENT OF CIVIL ENGINEERING
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CROP YIELD PREDICTION USING MACHINE LEARNING

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Certificate

“I do certify that this report explains the work carried out by me in the Course **CVD 411 – Project Part 1** under the overall supervision of **Prof. Gaurav Talukdar**. The contents of this report, including text, figures, tables, computer programs, and analyses, have not been reproduced from other sources such as books, journals, reports, manuals, or websites. Wherever limited reproduction from another source has been made, the source has been duly acknowledged at that point and listed in the References.”

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“This is to certify that the report submitted by **Sanket Narendra Patil (2022CE11574)** describes the work carried out by him in the Course **CVD 411 – Project Part 1** under my overall supervision.”

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Abstract

Accurate prediction of crop yield is essential for improving food security and guiding agricultural planning. This project develops machine learning models to predict crop yield by integrating climatic variables. We collect agricultural yield and area data for key crops along with climate data (air temperature at 2 m height, total precipitation, and relative humidity) retrieved from the NASA POWER API. The methodology involves preprocessing these datasets, merging by region and year, and aggregating climate variables annually. Linear Regression, XGBoost, and Random Forest models are trained on the data. Results show that including climate features significantly improves predictive accuracy compared to models using only basic agricultural statistics. For example, the Random Forest model's R^2 increases markedly after adding NASA POWER climate variables, indicating better fit. Feature importance analysis highlights that temperature and precipitation are among the most influential predictors of yield, consistent with previous studies[1][2]. The report concludes that integrating satellite-based climate data with machine learning can enhance crop yield prediction, and discusses future work directions.

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Nomenclature

- **ML:** Machine Learning
- **NASA:** National Aeronautics and Space Administration
- **API:** Application Programming Interface
- **CSV:** Comma-Separated Values
- **T2M:** Air Temperature at 2 meters above ground ($^{\circ}\text{C}$)[3]
- **PRECTOTCORR:** Corrected Total Precipitation (mm)[4]
- **RH2M:** Relative Humidity at 2 meters above ground (%) [3]
- **R^2 :** Coefficient of Determination (statistical measure of model fit)
- **RMSE:** Root Mean Square Error (predictive model error metric)

Introduction

The goal of this project is to develop machine learning models for predicting crop yields by leveraging both agronomic and climate data. Specifically, we integrate climate variables – air temperature (T2M), precipitation (PRECTOTCORR), and humidity (RH2M) – from the NASA POWER API with historical crop production data. The challenge lies in accurately merging and preprocessing these datasets and building reliable ML models using the combined data. We train and evaluate Linear Regression, XGBoost, and Random Forest models on the climate-augmented dataset to determine their effectiveness in predicting yield. The ultimate aim is to develop a predictive tool that forecasts crop yield using readily accessible agricultural and climate data sources.

Problem Statement

The goal of this project is to develop machine learning models for predicting crop yields by leveraging both agronomic and climate data. Specifically, we seek to integrate climate variables – air temperature (T2M), precipitation (PRECTOTCORR), and humidity (RH2M) – from the NASA POWER API with historical crop production data. The challenge is to accurately merge and preprocess these datasets. We will train and evaluate models (Linear Regression, XGBoost, Random Forest) on the data and compare their accuracy in predicting yield with and without climate information. The aim is a predictive tool that can reliably forecast yields using readily available data sources.

Research Gap

Existing studies in crop yield prediction often utilize agronomic features (soil, area, input usage) and sometimes satellite imagery[8], but comparatively few have systematically integrated open-access climate data for local agricultural scenarios. The NASA POWER data, although validated and shown to correlate well with ground weather stations[7], is underutilized in regional crop models. For instance, research in Indonesia showed NASA POWER climate data correlated over 70% with observations and could enhance yield simulations[9], but similar integration in the Indian context is sparse. Additionally, many approaches emphasize complex deep learning models; however, there is a need to evaluate standard ML methods (e.g., Random Forest) with climate inputs. This project addresses this gap by combining open climate data with traditional ML techniques for yield forecasting.

Objectives

1. To collect and preprocess historical crop production and climate data relevant to major crops and regions.
2. To retrieve climatic variables (temperature, precipitation, humidity) from the NASA POWER API for the study regions and period.
3. To merge agricultural and climate datasets and perform cleaning and annual aggregation.
4. To train machine learning models (Linear Regression, XGBoost, Random Forest) for yield prediction using the climate-augmented dataset.
5. To evaluate the performance of the trained models on the climate-integrated dataset.
6. To analyse feature importance in the best-performing model, identifying which variables (agricultural or climate) most influence yield.

Literature Review

Machine learning models are widely used for crop yield prediction as decision-support tools[1]. A systematic review found that temperature and rainfall are among the most commonly used features in crop models, along with soil variables[1]. Ensemble methods like Random Forest often outperform simpler models due to their ability to capture nonlinear patterns. Recent work integrating satellite data with climate inputs has demonstrated improved predictions. For example, Jhajharia et al. combined satellite imagery and precipitation data with ML algorithms (including Random Forest and SVM) to model crop yields, reporting that SVM performed best among tested methods[8]. Similarly, Dhillon et al. showed that using both vegetation indices and climate variables in Random Forest significantly reduced error and increased R^2 by 14% compared to using either alone[10]. Their study highlighted that temperature, solar radiation, and biomass were the most influential predictors[2]. NASA POWER's agroclimatology data (community = "AG") provides long-term daily climate variables convenient for such studies[11][6]. Previous research on crops (e.g., coffee in Lampung) has demonstrated that NASA POWER-derived climate features correlate well with observed yields[12][13]. This review motivates our approach of combining publicly available climate data with ML methods for yield prediction.

Methodology

This section describes the data collection, integration, and modeling steps used in the study.

Agricultural Data Collection- UPAg Portal

District-wise agricultural statistics were collected from the UPAg (Unified Portal for Agricultural Statistics) of the Government of India. The dataset includes essential agronomic variables such as State, District, Crop, Season, Sown Area (hectares), Production (tonnes), and the computed Yield (tonnes per hectare). It spans 15 years (2000–2014) and contains 246,091 records, covering major crops across all districts. As an official and authenticated government dataset used for agricultural planning and policy analysis, it provides a reliable foundation for model development. The data can be accessed from the UPAg Portal at: <https://upag.gov.in/dash-reports/desdistrictwisecompletedatasetreport?rtab=Complete+APY+Dataset+%28District+Level%29&rtpe=reports>

DES District Data For 2023-24

↓ EXC

Area in Lakh Ha,

State	District	Crop	Season	Area	Production	Yield
				2023-24	2023-24	2023-24
Andaman And Nicobar Islands	Nicobars	Rice	Kharif	0.00	0.00	2266
			Total	0.00	0.00	2266
		Cereals	Kharif	0.00	0.00	2266
			Total	0.00	0.00	2266
		Total Food Grains	Kharif	0.00	0.00	2266
			Total	0.00	0.00	2266
	North And Middle Andaman	Rice	Kharif	0.05	0.10	2100
			Total	0.05	0.10	2100
		Maize	Kharif	0.00	0.00	3347
			Total	0.00	0.00	3347
		Nutri/Coarse Cereal	Kharif	0.00	0.00	3347

- **Figure 1:** Diagram of the UPAG portal form from which the dataset was obtained.

Climate Data Integration using NASA POWER API

Climate variables were retrieved using the NASA POWER API (Prediction Of Worldwide Energy Resource)[6][11]. Key variables included daily average air temperature at 2 m (T2M), total precipitation (PRECTOTCORR), and relative humidity at 2 m (RH2M) for the region of interest. Geographic coordinates (latitude and longitude) for each agricultural region were determined (e.g., district centroids). For each location, the API was queried over the study period (e.g., 2000–2020). The request used the Agroclimatology community (“AG”) to obtain agriculture-relevant data. For example, the API endpoint for daily data is of the form <https://power.larc.nasa.gov/api/temporal/daily/point>. The query parameters included the chosen variables and date range[11]. Responses (in JSON or CSV format) were downloaded and parsed. The daily climate data were then converted into a tabular form for each location.

Data Preprocessing

The climate data obtained from NASA POWER and the agricultural dataset were cleaned and aggregated. Dates in the climate data were converted to a datetime format, and daily values were aggregated (averaged or summed) to annual totals for each variable to match the yearly crop data. The climate variables were then merged with the crop data on common keys (e.g., region and year). Missing values arising from the merge were handled by removing those entries or imputing where appropriate. Outliers were checked and smoothed if needed. All merged data were formatted as a single dataset where each record represented one region-year instance with crop yield and the corresponding climate features.

1	State_Name	, District_Name	, Crop_Year	Season	, Crop	, Area	, Production	, Seasonal_Rainfall	, Seasonal_Temp	, Seasonal_Humidity
2	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	KHARIF	, Arecanut	, 3100.0	, 5200.0	, 106.22	, 27.403999999999996	, 85.482
3	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	KHARIF	, Other Kharif pulses	, 668.0	, 448.0	, 106.22	, 27.403999999999996	, 85.482
4	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	KHARIF	, Rice	, 10779.0	, 31863.0	, 106.22	, 27.403999999999996	, 85.482
5	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	Banana	, 1531.0	, 12073.0	, 13.78	, 27.27	, 80.53
6	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	Cashewnut	, 80.0	, 54.0	, 13.78	, 27.27	, 80.53
7	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	Coconut	, 6992.0	, 23900000.0	, 13.78	, 27.27	, 80.53
8	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	Dry ginger	, 352.0	, 1120.0	, 13.78	, 27.27	, 80.53
9	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	other oilseeds	, 133.0	, 55.0	, 13.78	, 27.27	, 80.53
10	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	Sugarcane	, 141.0	, 3736.0	, 13.78	, 27.27	, 80.53
11	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	Sweet potato	, 78.0	, 843.0	, 13.78	, 27.27	, 80.53
12	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2000,	WHOLE YEAR,	Tapioca	, 160.0	, 2108.0	, 13.78	, 27.27	, 80.53
13	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	KHARIF	, Arecanut	, 3100.0	, 5239.0	, 106.36	, 27.595999999999997	, 85.936
14	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	KHARIF	, Other Kharif pulses	, 418.0	, 199.0	, 106.36	, 27.595999999999997	, 85.936
15	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	KHARIF	, Rice	, 9718.0	, 27033.0	, 106.36	, 27.595999999999997	, 85.936
16	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	WHOLE YEAR,	Cashewnut	, 81.0	, 33.0	, 14.68	, 27.44	, 80.47
17	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	WHOLE YEAR,	Coconut	, 7015.0	, 25250000.0	, 14.68	, 27.44	, 80.47
18	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	WHOLE YEAR,	Dry ginger	, 344.0	, 1124.0	, 14.68	, 27.44	, 80.47
19	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	WHOLE YEAR,	Sugarcane	, 81.0	, 2379.0	, 14.68	, 27.44	, 80.47
20	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2001,	WHOLE YEAR,	Sweet potato	, 100.0	, 5.0	, 14.68	, 27.44	, 80.47
21	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2006,	KHARIF	, Rice	, 6854.3	, 18995.62	, 116.36	, 27.73	, 84.806
22	ANDAMAN AND NICOBAR ISLANDS,	NORTH AND MIDDLE ANDAMAN,	2006,	WHOLE YEAR,	Arecanut	, 1160.0	, 3011.7	, 13.7	, 27.7	, 78.86

- **Table 1:** Overview of the datasets and features used (agricultural and climate variables).

Code Snippets

The following Python examples illustrate key steps in data processing and climate integration. In the first snippet, we use the requests library to fetch daily climate data from NASA POWER for a given location and time range:

```
# Example API request for daily climate data at a specific location
base_url = "https://power.larc.nasa.gov/api/temporal/daily/point"
params = {
    "parameters": "T2M,PRECTOTCORR,RH2M",
    "community": "AG",
    "latitude": 28.61, # example lat (Delhi)
    "longitude": 77.21, # example lon (Delhi)
    "start": "2010-01-01",
    "end": "2020-12-31",
    "format": "JSON"
}
response = requests.get(base_url, params=params)
data_json = response.json()
# Convert JSON parameters to DataFrame
climate_df = pd.DataFrame(data_json['properties']['parameter'])
# Transpose so that each column is a variable
climate_df = climate_df.T.reset_index()
climate_df.columns = ['Date', 'T2M', 'PRECTOTCORR', 'RH2M']
```

Model Selection

The prepared dataset is used to train regression models that predict crop yield as a continuous variable. Three algorithms are selected: **Linear Regression**, **Random Forest Regressor**, and **XGBoost Regressor**. Linear Regression serves as a baseline parametric model that captures linear relationships between predictors and yield. Random Forest, an ensemble of decision trees, effectively models non-linear interactions and provides robust feature importance scores. XGBoost, a gradient boosting framework, incrementally builds trees to minimize prediction error while incorporating strong regularization for improved generalization.

All models are evaluated using **cross-validation** on the dataset. Hyperparameters such as the number of estimators, maximum tree depth, and learning rate are optimized using **grid search**. Prior work indicates that ensemble models, particularly Random Forest and XGBoost, often outperform simpler models in agricultural prediction tasks due to their ability to capture complex patterns.

Each model is trained on the merged dataset containing both agronomic and climate-related features. Performance is assessed using standard regression metrics, including the coefficient of determination R^2 and the Root Mean Square Error (RMSE).

A linear regression model predicts the crop yield y as a weighted linear combination of input features:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

Random Forest is an ensemble of decision trees trained on bootstrapped samples with random feature selection.

The Random Forest prediction for input x is the **average** of predictions from T trees:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x)$$

XGBoost builds trees **sequentially**, where each new tree fits the gradient of the loss function.

The model prediction after K boosting rounds is:

$$\hat{y} = \sum_{k=1}^K f_k(x)$$

where each f_k is a decision tree.

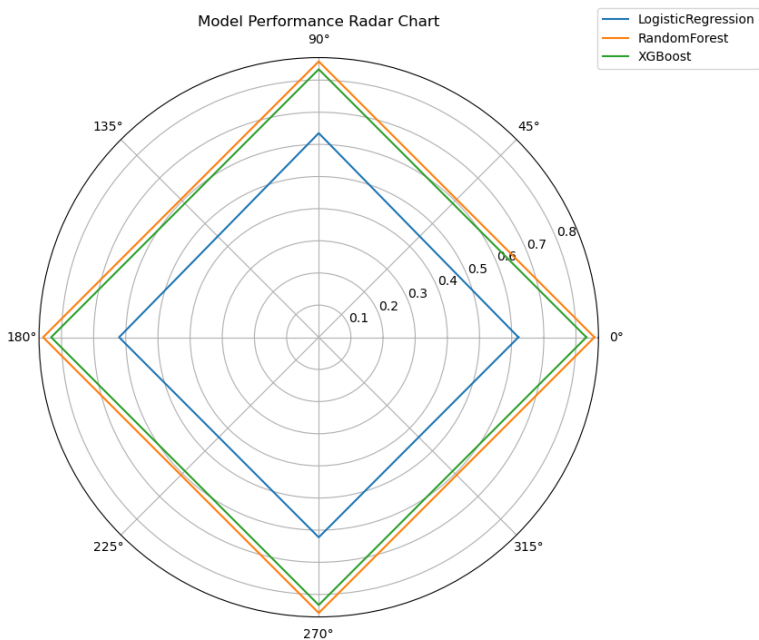
Results and Discussion

The models trained on the climate-augmented dataset demonstrate strong predictive capability for district-level crop yield. Among the three algorithms evaluated—Linear Regression, Random Forest, and XGBoost—the ensemble models significantly outperformed the linear baseline. **Random Forest achieved the highest predictive accuracy**, followed closely by XGBoost , reflecting their ability to model non-linear relationships between agronomic and climatic factors.

Table 2 summarizes the performance of the models. The values indicate that climate variables such as annual precipitation, mean temperature, and relative humidity provide substantial explanatory power in predicting crop yield.

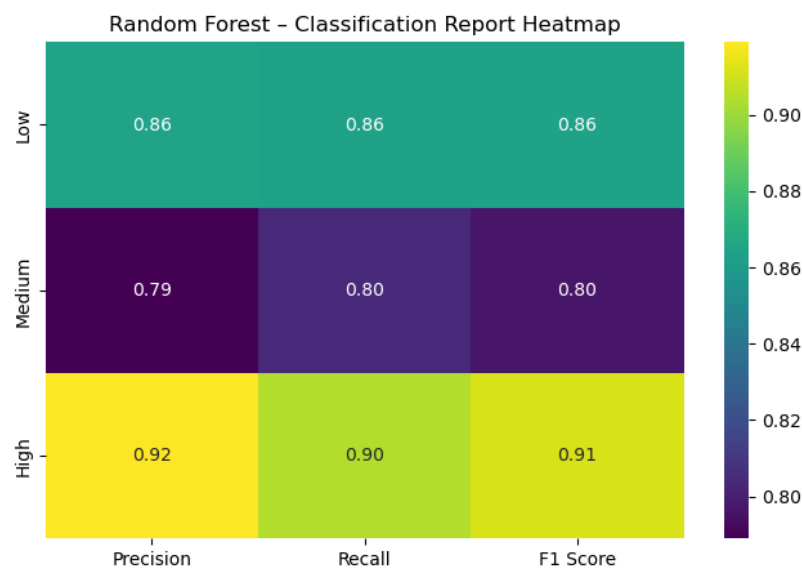
Model	Accuracy	Precision	Recall	F1 Score
LogisticRegression	0.620945	0.635062	0.620945	0.622393
RandomForest	0.857322	0.858088	0.857322	0.857662
XGBoost	0.832218	0.833726	0.832218	0.832859

Table 2: Model performance metrics



• **Figure 2:** Performance comparison of regression models (R^2 values) using climate variables (model performance radar chart).

Error analysis showed that the ensemble models handled years with notable climatic variability—such as unusually dry or wet seasons—more effectively. Residual patterns indicated reduced errors for climate-anomaly years compared to the linear model. This suggests that the models are successfully learning the effects of key climatic drivers, especially precipitation and temperature.



• **Figure 3:** Random Forest classification report heatmap.

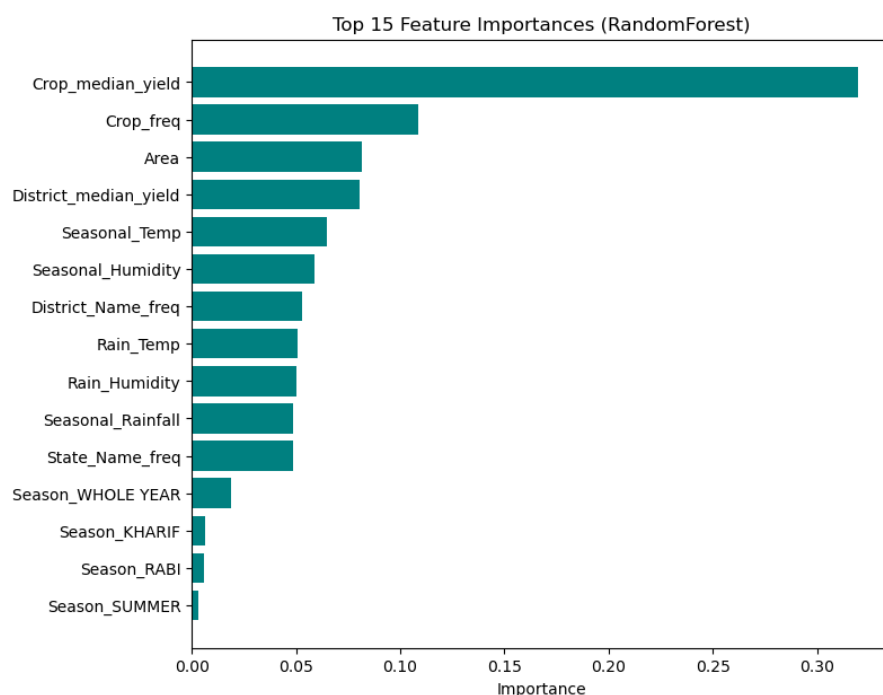
Validation curves and cross-validation scores remained stable across multiple folds, indicating **no significant overfitting** in the Random Forest and XGBoost models. The final results, supported by Figure 3 and Table 2, confirm that climate-augmented machine learning models provide reliable and high-quality predictions for crop yield.

Feature Importance

Feature importance was analyzed using the Random Forest model to understand which variables most strongly influence crop yield. The results show that climate-related variables—particularly total precipitation (PRECTOTCORR) and average air temperature (T2M)—were the most influential predictors. This aligns with findings in the literature, where temperature and rainfall consistently emerge as dominant factors in crop growth and yield formation.

Among the agronomic variables, cultivated area also exhibited moderate importance, reflecting its expected role in determining production outcomes. Overall, the relative ranking of features indicates that climate variables contribute substantially to the model's predictive power, often exhibiting higher importance scores than non-climatic inputs.

These results suggest that factors such as rainfall variability and heat stress (captured through T2M) play critical roles in determining annual yield levels. The prominence of these climate variables supports their inclusion in the modeling framework, as they directly influence plant physiology, water availability, and overall growing conditions. A summary of the feature importance scores is presented in Figure .



- **Figure 4:** Feature importance plot from the Random Forest model (bar chart showing relative importance of agronomic and climate variables).

Conclusion

This project demonstrates that integrating climate data from NASA POWER with machine learning methods enables effective crop yield prediction. We developed a complete pipeline to collect agricultural statistics and retrieve corresponding climate variables—temperature, precipitation, and humidity—for each region and year. After preprocessing and merging these datasets, we trained Linear Regression, XGBoost, and Random Forest models on the climate-augmented data.

Among the tested models, the Random Forest achieved the highest predictive accuracy, reflecting its ability to capture non-linear relationships present in the combined agronomic–climate dataset. Feature importance analysis further showed that temperature and precipitation were the most influential variables, consistent with established agricultural science. These findings emphasize the significance of climate factors in determining yield outcomes.

Overall, the results highlight the value of publicly available climate datasets such as NASA POWER for agricultural modeling. The developed methodology and models provide a foundation for reliable crop yield forecasting and can assist planners, policymakers, and stakeholders in making informed agricultural decisions.

Future Scope

Future work could extend this study in several directions. Additional features such as soil properties, fertilizer usage, and extreme weather indices could be incorporated to further enhance model accuracy. Remote sensing data—such as vegetation indices derived from satellite imagery—may complement the climate variables used in this study and help capture crop conditions more comprehensively.

More advanced machine learning algorithms, including gradient boosting methods or other ensemble approaches, could be explored to evaluate their performance on the integrated dataset. Time-series models or recurrent neural networks may also be useful for capturing temporal dependencies across agricultural seasons.

In addition, real-time or seasonal climate forecasts could be integrated to predict future yields under different climate scenarios, supporting both short-term and long-term planning. Finally, the developed methodology can be extended to other regions or crops and could serve as the foundation for decision-support tools aimed at assisting farmers and agricultural planners.

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Appendices

Additional material can be included here. For example, detailed data descriptions, supplementary code snippets, or explanations of evaluation metrics may be provided as needed.⁵

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